IDB 402 — Projects in Linguistics

Research Paper

A Computational Gender Analysis of Pride and Prejudice: Dependency Parsing and the Encoding of Gendered Agency in Austen's Syntax

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Abstract

This study presents a computational analysis of gender dynamics in Jane Austen's Pride and Prejudice using syntactic dependency parsing. The pipeline extracts subject–verb, object–verb, and adjective–noun relations across the novel's approximately 146,000 tokens to measure how male and female characters are linguistically constructed in terms of agency (acting versus being acted upon), characterization (attributive versus predicative adjectives), and participation in specific semantic frames such as marriage. Gender is assigned through a three-tier cascade (gendered pronoun → role noun → named entity), with context-aware surname disambiguation resolving ambiguous surnames against preceding title tokens. A preliminary sentiment-analysis component was dropped in favor of collocation and verb-frame analyses after domain-mismatch testing against 19th-century prose. Statistical significance is tested with Fisher's exact test, and Cramér's V is reported as the effect-size measure. Results surface an "agency paradox": female characters show a higher headline agent/patient ratio than male characters (0.8177 versus 0.7620), inverting the gap reported by Stuhler (2024) at large-corpus scale. The paradox dissolves under verb-level inspection, female agency concentrates in verbs of perception and cognition, while female referents are about 2.5× as likely as male referents to occupy the patient position within the marry frame. (Fisher's exact $p < 0.001$; Cramér's $V = 0.397$, a large effect). The study is grounded in feminist digital humanities (D'Ignazio & Klein, 2020) and argues that headline agency metrics must be read against narrative focalization rather than treated as standalone evidence of representation.

Keywords: computational stylistics, dependency parsing, collocation analysis, gender agency, Jane Austen, Pride and Prejudice, feminist digital humanities, corpus linguistics.

1. Introduction and Theoretical Background

It is a truth universally acknowledged, that Jane Austen's Pride and Prejudice (1813) has long been recognised as a novel deeply concerned with gender, social power, and the constraints placed on women's agency within Regency-era England. Traditional literary scholarship has explored these themes extensively through close reading, but computational methods now offer the possibility of testing received interpretive claims against measurable linguistic evidence. This project asks: can the syntactic structures of Austen's prose — the verbs characters perform, the adjectives that describe them, the grammatical roles they occupy — quantitatively reveal the gendered asymmetries that literary critics have long identified?

The computational study of Austen's language has a substantial pedigree. Burrows's Computation into Criticism (1987) demonstrated that even common function-word frequencies differentiate Austen's characters in ways that align with close-reading interpretations. Fischer-Starcke (2009, 2010) applied corpus-linguistic methods directly to Pride and Prejudice, revealing through keyword analysis that computational approaches surface social dynamics not previously foregrounded in traditional criticism. More recently, Bamman, Underwood, and Smith (2014) established the foundational methodology for representing literary characters through typed dependency relations — agent verbs, patient verbs, possessive nouns, and predicative adjectives — using a corpus of over 15,000 novels from HathiTrust. Baker's Using Corpora to Analyze Gender (2014) provides the methodological grounding for treating gendered collocational patterns as evidence of ideological structure rather than mere stylistic texture.

The most methodologically aligned precedents for this project are Stuhler (2024), who analysed 87,531 fiction works using dependency parsing to formalise agency as the ratio between agent and patient roles and reported a persistent "gender agency gap" in which female characters are portrayed as more passive; Jockers and Kirilloff (2016), who extracted pronoun–verb pairs from approximately 3,000 19th-century novels and found that female pronouns were strongly associated with emotional and stative verbs ("wept," "sat," "felt") while male pronouns clustered with action verbs ("took," "walked," "rode"); and Underwood, Bamman, and Lee (2018), who demonstrated that gender divisions in fiction were sharpest in the mid-19th century — precisely Austen's era. Cheng (2020) confirmed these patterns using spaCy specifically, finding that bodily language plays a consistently larger role in configuring fictional women than men.

The feminist digital humanities framework grounds the project's interpretive stance. D'Ignazio and Klein (2020) argue for examining and challenging power structures in data, elevating emotion as a legitimate analytical lens, and exercising caution about treating gender as a simple binary. Mandell (2019) warns that computational text mining risks reinforcing the very gender stereotypes it studies, recommending that computational findings serve as "signposts" for targeted close reading. Rhody (2016) calls for "productive discomfort", transparency about methodological tensions, and Risam (2018) urges consideration of what is lost when applying 21st-century NLP models to early 19th-century prose.

1.1. Aim and Scope

The aim of this study is to examine how gendered agency is syntactically encoded in Jane Austen's *Pride and Prejudice* through computational dependency parsing. More specifically, the study investigates whether male and female referents are distributed differently across agent and patient positions, whether particular verbs and adjectives are more strongly associated with one gender than the other, and whether specific semantic frames, especially the marriage frame, reveal gendered patterns that may be obscured by aggregate agency measures. By combining corpus-linguistic methods with insights from feminist digital humanities and Critical Discourse Analysis, the study aims to show how computational evidence can contribute to the interpretation of gender, power, and social positioning in a canonical literary text.

The scope of the study is limited to the full text of *Pride and Prejudice*, sourced from Project Gutenberg, and to syntactic relations that can be operationalised through dependency parsing, namely subject–verb, object–verb, adjective–noun, predicative adjective, and possessive relations. Gender attribution is restricted to male and female referents identifiable through pronouns, gendered role nouns, and named entities. The analysis focuses on grammatical agency, character description, collocation profiles, and selected verb frames rather than on all possible aspects of gender representation in the novel.

This study does not attempt to provide a complete literary interpretation of *Pride and Prejudice*, nor does it generalise its findings to all of Austen's novels or to nineteenth-century fiction as a whole. It also does not include reader reception, historical biography, or a full manual close reading of every relevant passage. A preliminary sentiment-analysis component was excluded from the final scope because off-the-shelf sentiment models proved unsuitable for Austen's irony-heavy and historically distant prose. Therefore, the study should be understood as a focused computational-linguistic investigation of gendered syntactic patterning in one novel, with its findings treated as evidence for targeted interpretation rather than as exhaustive claims about Austen's entire representation of gender.

2. Research Questions

The revised study is organised around four research questions:

1. How do the syntactic roles (agent versus patient) of male and female characters in Pride and Prejudice differ, and what do these differences reveal about the encoding of gendered agency in Austen's prose?
2. Which verbs and adjectives are most distinctively associated with male and female characters through dependency relations (nsubj, dobj, amod, acomp), and do these patterns align with or challenge established findings from large-scale corpus studies (Jockers & Kirilloff, 2016; Stuhler, 2024; Underwood et al., 2018)?

3. How does a targeted semantic frame — most prominently the marriage frame — distribute male and female characters between agent and patient positions, and how strong is any observed asymmetry?
4. Do headline agency metrics survive decomposition by verb class and semantic frame, or does narrative focalisation through a single protagonist invert them at the single-text scale?

3. Methodology

3.1. Corpus and Preprocessing

The full text of Pride and Prejudice is sourced from Project Gutenberg (EBook #1342). Gutenberg headers and footers are stripped manually, then transformed into CSV by corpus_builder.py. The cleaned corpus is tokenised, lemmatised, and dependency-parsed using spaCy's transformer-based en_core_web_trf model, producing a DataFrame of approximately 146,000 tokens with dependency labels, part-of-speech tags, and lemmatised forms. The transformer model was chosen over en_core_web_sm after pilot testing showed meaningfully higher parsing accuracy on Austen's syntactically complex, clause-embedded sentences.

3.2. Gender Attribution

Gender is assigned to referring expressions through a three-tier cascade: (1) gendered pronouns, (2) gendered role nouns (Mr., Mrs., Miss, Lady, Sir, brother/sister and their plural and possessive forms), and (3) named entities. Ambiguous surnames — most conspicuously Bennet and Bingley, which are shared across several characters of both genders — are resolved against the preceding title token (Mr./Mrs./Miss) within a small left-context window. This context-aware disambiguation is the single most consequential difference between the revised pipeline and the version described in the initial proposal.

3.3. Dependency-Based Agency and Description

Following Stuhler's (2022, 2024) entity-centred semantic grammar, characters are classified into four syntactic roles based on their dependency relations:

- Agent: characters appearing as nominal subjects (nsubj) of active verbs, representing acting entities.
- Patient: characters appearing as direct objects (dobj) of transitive verbs, representing entities acted upon.
- Described: characters modified by adjectives, with attributive modifiers (amod) and predicative complements (acomp) treated separately so that physical/demographic description is not collapsed into psychological/evaluative characterisation.
- Possessor: characters in possessive constructions (poss), capturing what characters own or are associated with.

The agency ratio is computed per gender as agent / (agent + patient), allowing direct comparison against the single-text scale reported in large-corpus studies.

3.4. Verb-Frame and Collocation Analysis

A preliminary two-model sentiment ensemble (VADER with Austen-specific patches plus a transformer classifier) was implemented during the pilot phase but has been dropped from the revised pipeline. Off-the-shelf sentiment models mismatch 19th-century prose — free indirect discourse, archaic intensifiers, and irony-heavy narration — in ways that produced inconclusive results and would have required disproportionate justification for a traditional-linguistics audience; this mirrors concerns raised in Rebora's (2023) survey of sentiment analysis in literary studies. The revised pipeline substitutes:

- Per-gender verb frequency distributions, normalised per 1,000 tokens, to identify verbs distinctively associated with male or female characters.
- Per-gender adjective distributions, split by dependency type (amod versus acomp), to distinguish how characters are presented attributively from how they are evaluated predicatively.
- Targeted semantic-frame analysis for selected verbs — most prominently marry — to quantify the agent/patient asymmetry within a single predicate rather than across the aggregate.

3.5. Statistical Testing

Verb-frame contingency tables are sparse, so Fisher's exact test is applied rather than chi-square. Cramér's V is reported alongside each p-value as a scale-independent effect-size measure, since p-values on a corpus of this size are uninformative about magnitude.

3.6. Dialogue/Narration Separation

Unicode curly quotation marks (U+201C and U+201D) are used to partition the text into narrated and spoken strands. This allows patterns in Austen's narrative voice to be analysed separately from patterns in her characters' speech, a distinction that matters for the free-indirect-discourse-heavy passages where focalisation and voice interact.

4. Data Analysis

The full pipeline has produced a first authoritative set of quantitative results. This section walks through six analytical slices — headline agency ratios, agency decomposed by verb class, the marriage frame, patient positioning and possessives, collocation profiles, and adjectival characterisation — and reads each against the scholarly precedents cited in Section 1.

4.1. Headline Agency Ratios

Agency is operationalised as the proportion of a character's tracked dependency relations that occupy the nsubj position of an active verb, against the sum of agent and patient (dobj) occurrences. The aggregate ratios are 0.8177 for female characters and 0.7620 for male characters — a result which, read in isolation, would suggest that Pride and Prejudice grants women more syntactic agency than men.

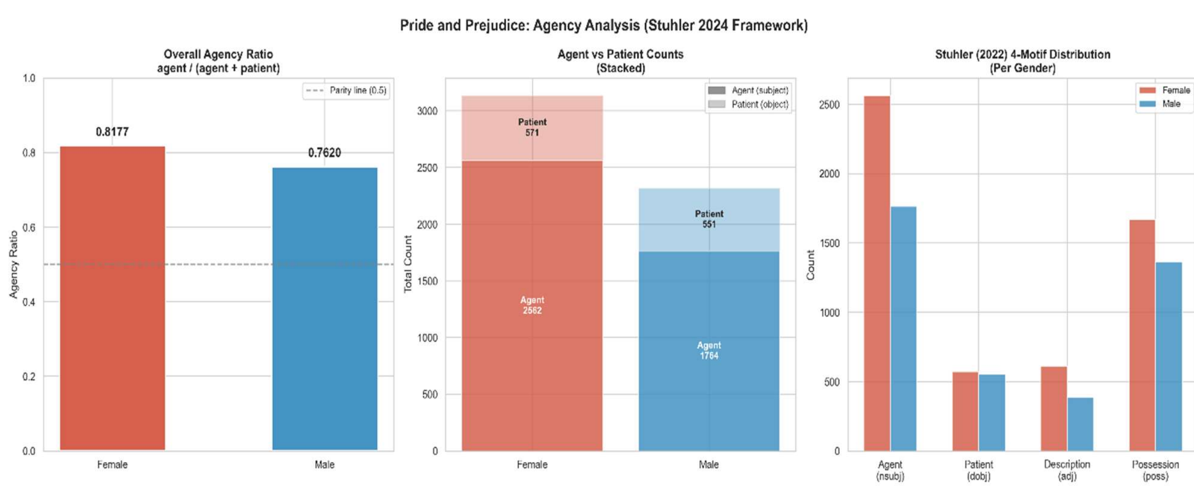

Figure 1. *Aggregate agency ratios (agent / (agent + patient)) computed per gender across the full novel. Female ratio 0.8177; male ratio 0.7620.*

This inverts the direction of the gap reported by Stuhler (2024) for a corpus of 87,531 fiction works and differs from the pattern Jockers and Kirilloff (2016) identified across 3,000 19th-century novels. A tier-level decomposition confirms that the inversion is not a fluke of any single gender-tier: pronouns, role nouns, and named entities all show the same direction.

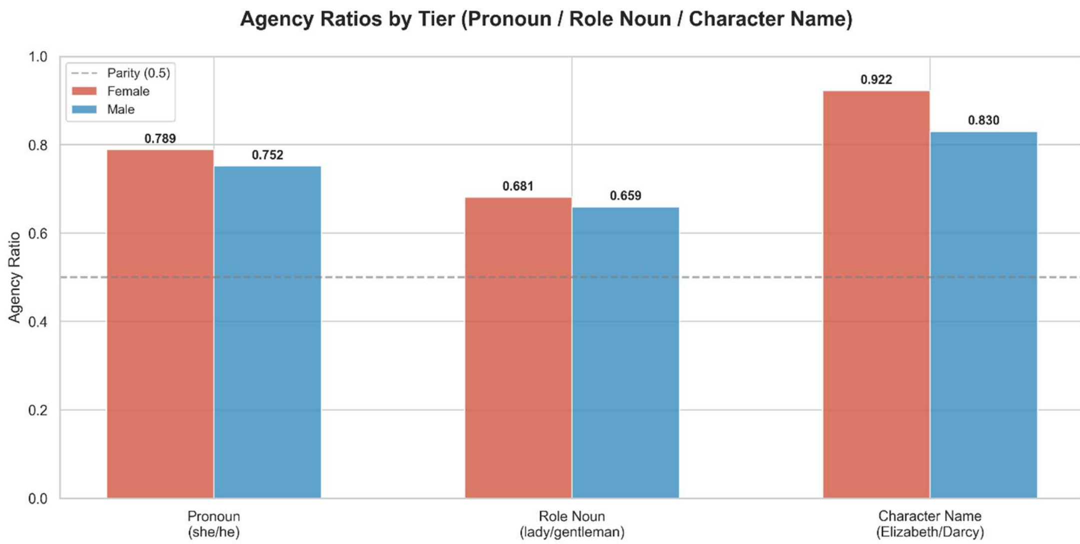

Figure 2. *Agency ratios decomposed across the three-tier gender cascade (gendered pronoun → role noun → named entity). The headline inversion is visible at every tier.*

The interpretive claim here is deliberately modest: the ratios are what they are, but a ratio is only meaningful once its constituent verbs are examined. Sections 4.2 and 4.3 do exactly that.

4.2. Agency Decomposed by Verb Class

When the per-gender agent counts are broken down by the verbs they attach to, the higher female ratio resolves into a clear semantic pattern. Female subjects cluster around verbs of perception, cognition, and interior state — feel, think, hope, wonder, believe — while male subjects cluster around verbs of motion, communication, and overt action — come, go, say, take, walk. This pattern echoes Jockers and Kirilloff (2016) and Cheng (2020), but surfaces at single-novel scale.

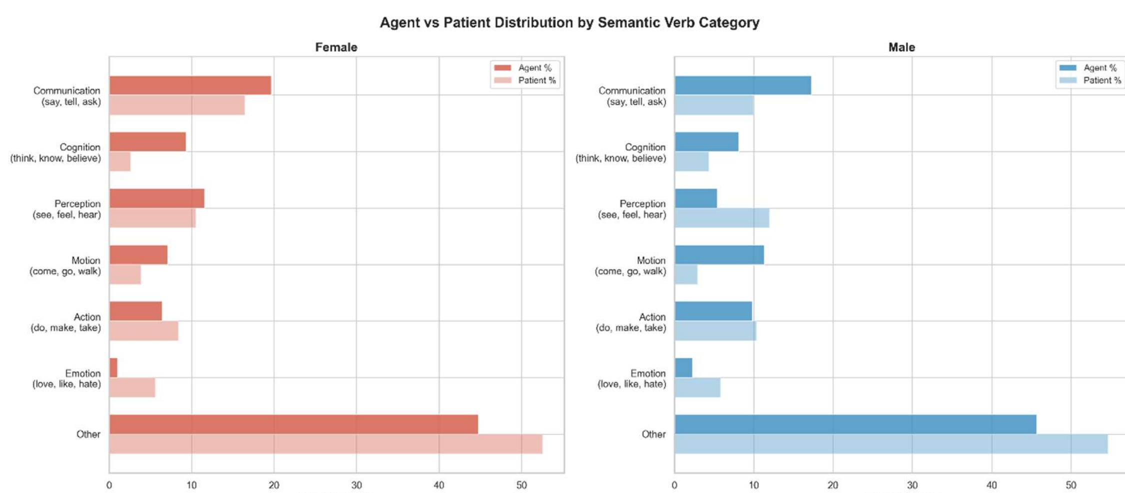

Figure 3. *Agent/patient distribution by semantic verb category. Female agency is more concentrated in cognition and perception, while male agency is more concentrated in communication, motion, and action.*

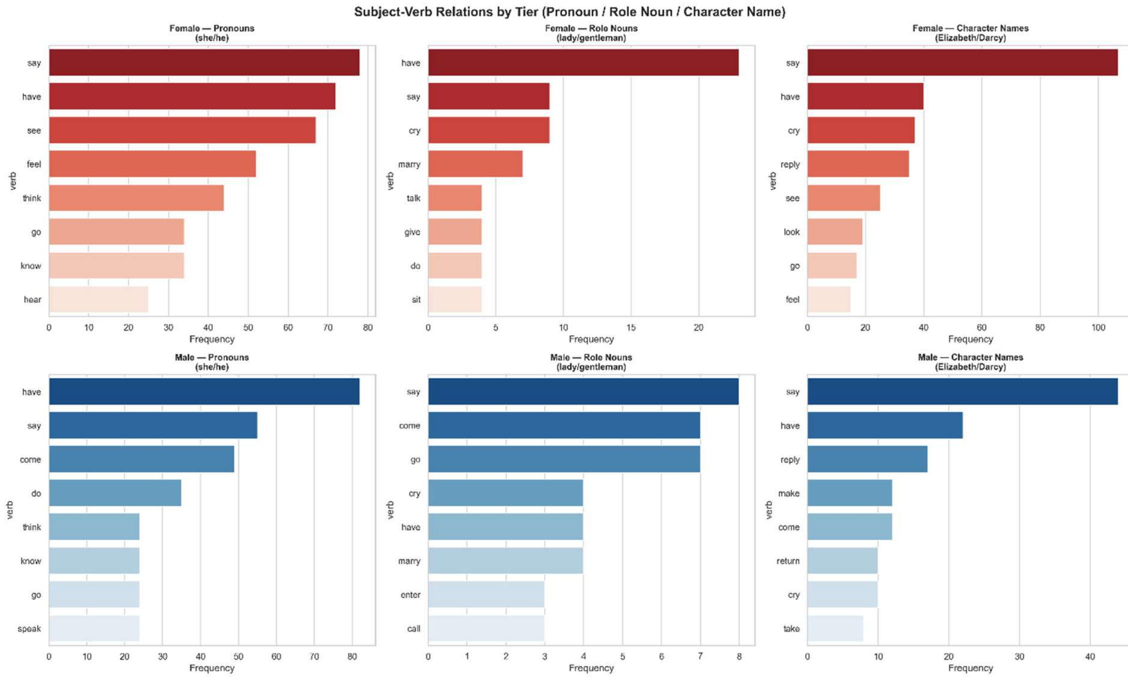

Figure 4. *Verb profile decomposed across the three gender-attribution tiers, confirming that the clustering is not an artefact of pronoun dominance.*

The reading that naturally follows is a familiar one from feminist narratology: Elizabeth Bennet is the novel's focalising consciousness, so verbs of perception and cognition accumulate around female subjects because Elizabeth perceives and reflects at length. Higher female agency, in this corpus, is a symptom of focalisation, not of empowerment.

4.3. The Marriage Frame

To test whether the focalisation reading holds up, the pipeline restricts attention to a single verb whose patient position is where gender ideology would most visibly inscribe itself: marry. Female referents are about 2.5× as likely as male referents to occupy the patient position within the *marry* frame. A Fisher's exact test rejects independence at $p < 0.001$, and Cramér's $V = 0.397$ places the effect in the large range for a 2×2 contingency table.

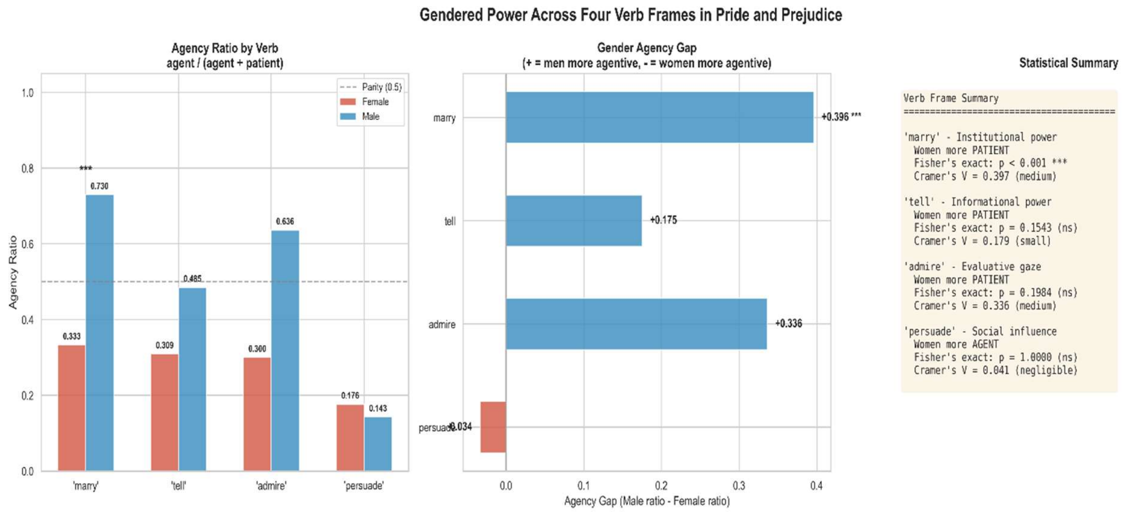

Figure 5. *Summary of agency ratios, agency gaps, and statistical results across the four tested verb frames: marry, tell, admire, and persuade.*

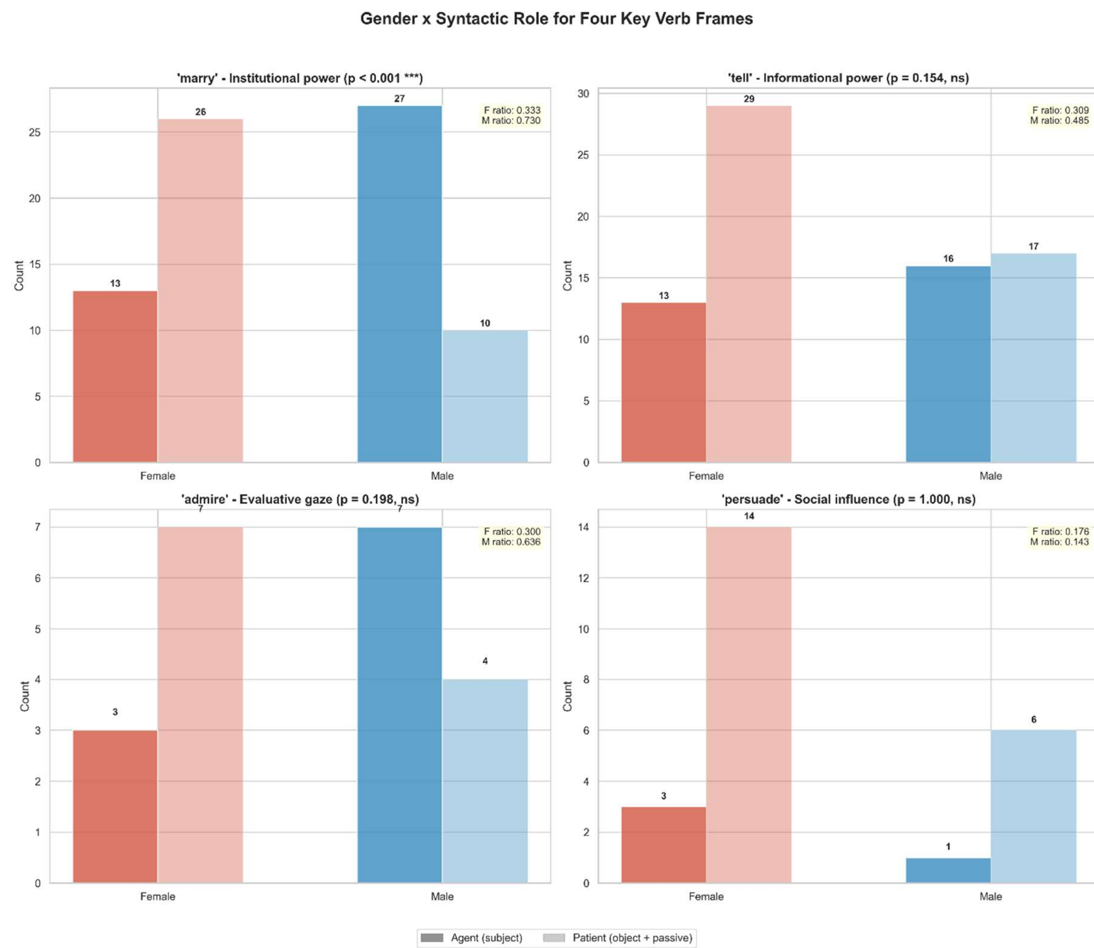

Figure 6. *Gender × syntactic-role distributions for the four tested verb frames: marry, tell, admire, and persuade.*

This is the strongest statistical finding in the study and the most compelling single piece of evidence that gender ideology is encoded at the syntactic level. The aggregate agency ratio in Section 4.1 tells a misleading story; the marriage frame, meanwhile, corrects it. Women are not absent as agents of matrimony, but they are much more frequently positioned as patients within the *marry* frame. Austen's grammar therefore represents marriage as a frame in which female referents are disproportionately acted upon rather than acting.

4.4. Patient Positioning and Possessives

Beyond the marriage frame, the broader patient-role distribution shows female referents frequently appearing as objects of high-frequency social and perceptual verbs such as *see*, *make*, *marry*, *assure*, *ask*, and *find*. These patterns suggest that female characters are often positioned as objects of perception, judgement, social action, and interpersonal negotiation. The possessive profiles add a relational dimension to this pattern. Female possessive forms most frequently modify kinship and relational nouns such as *sister*, *mother*, *daughter*, *father*, *friend*, and *husband*, while male possessives include both relational nouns and broader social-evaluative heads such as *character*, *behaviour*, *manner*, and *attention*. This suggests that possession does not simply indicate ownership but also encodes social relationality.

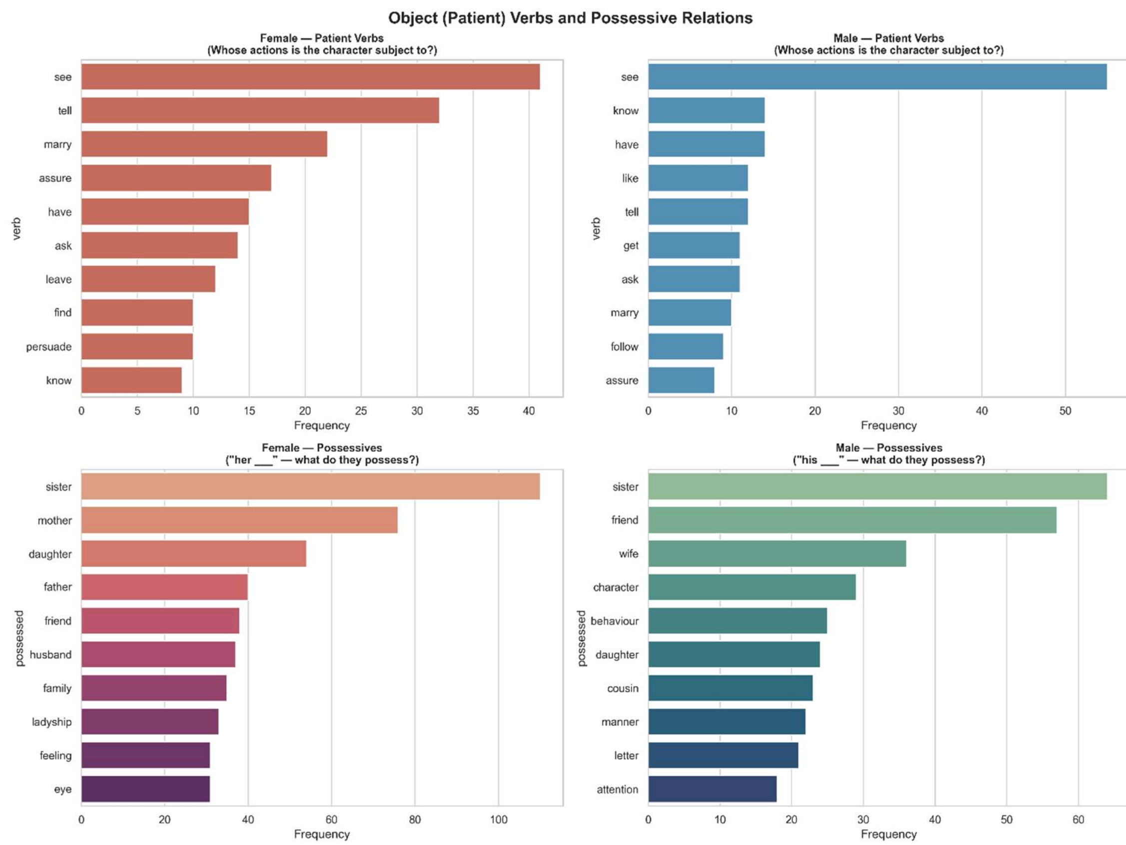

Figure 7. *Patient-role verbs and top possessive contexts by gender*

4.5. Collocation Profiles

The dropped sentiment component is compensated for by a collocation view, following the methodological grounding Baker (2014) provides for corpus-linguistic gender analysis. Normalised per 1,000 gendered tokens, *feel* appears several times more often for female referents, while *come* appears roughly three to four times more often for male referents — a quantitative restatement of the Jockers and Kirilloff (2016) result, holding at single-novel scale.

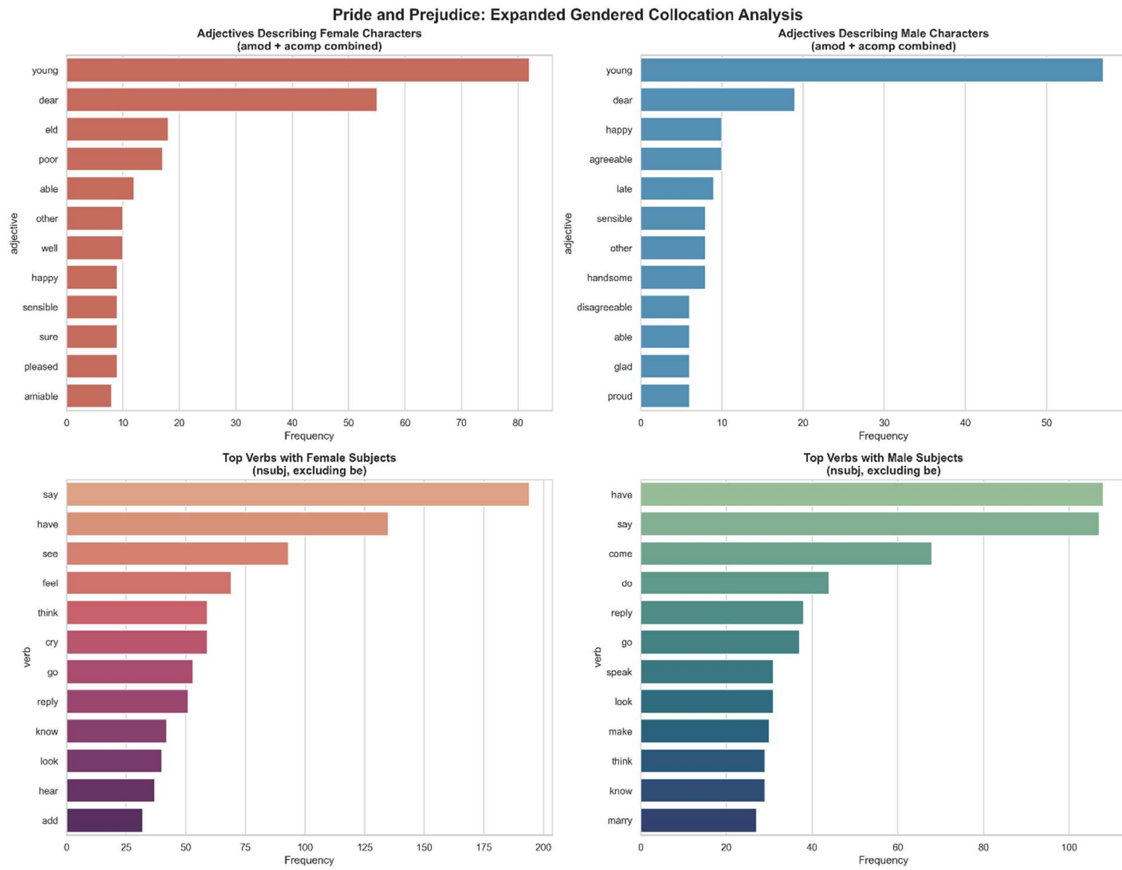

Figure 8. *Raw frequency profiles for adjectives and subject verbs by gender.*

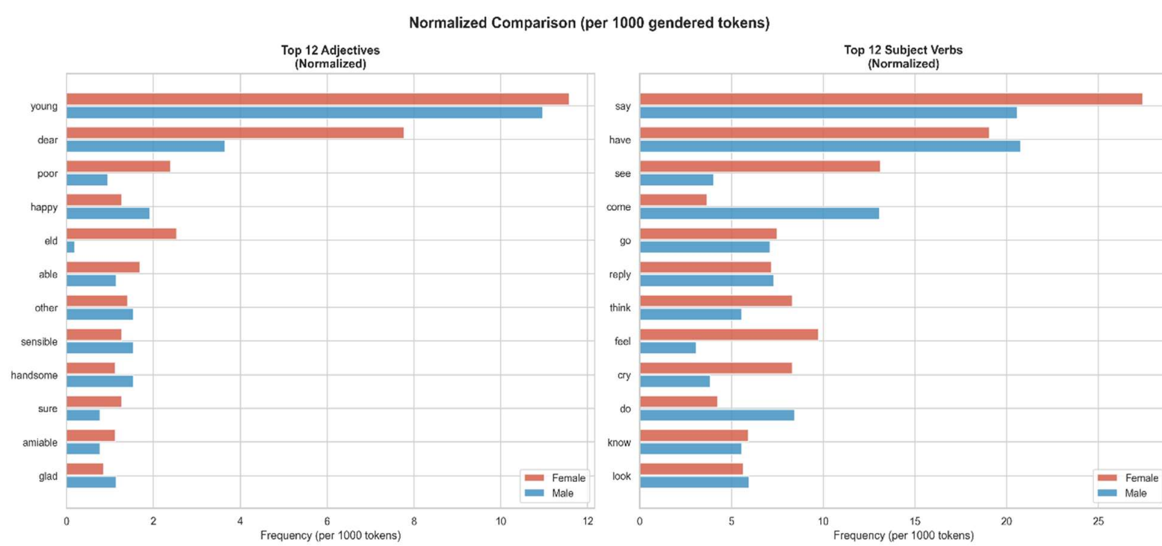

Figure 9. *Normalised adjective and subject-verb frequencies per 1,000 gendered tokens.*

4.6. Adjectival Characterisation

Finally, attributive modifiers (amod) and predicative complements (acomp) are separated so that the two kinds of description do not cancel each other out in the aggregate. Attributive adjectives attaching to female referents are dominated by physical, age-related, evaluative and relational descriptors such as *young*, *dear*, *old*, and *poor*. Predicative adjectives, by contrast, more often encode evaluative or psychological states, including *able*, *well*, *sure*, *pleased*, *sorry*, *happy*, and *glad*. The male adjective profile is less clearly reducible to rank or estate, but it similarly mixes social evaluation, affective state, and character judgement.

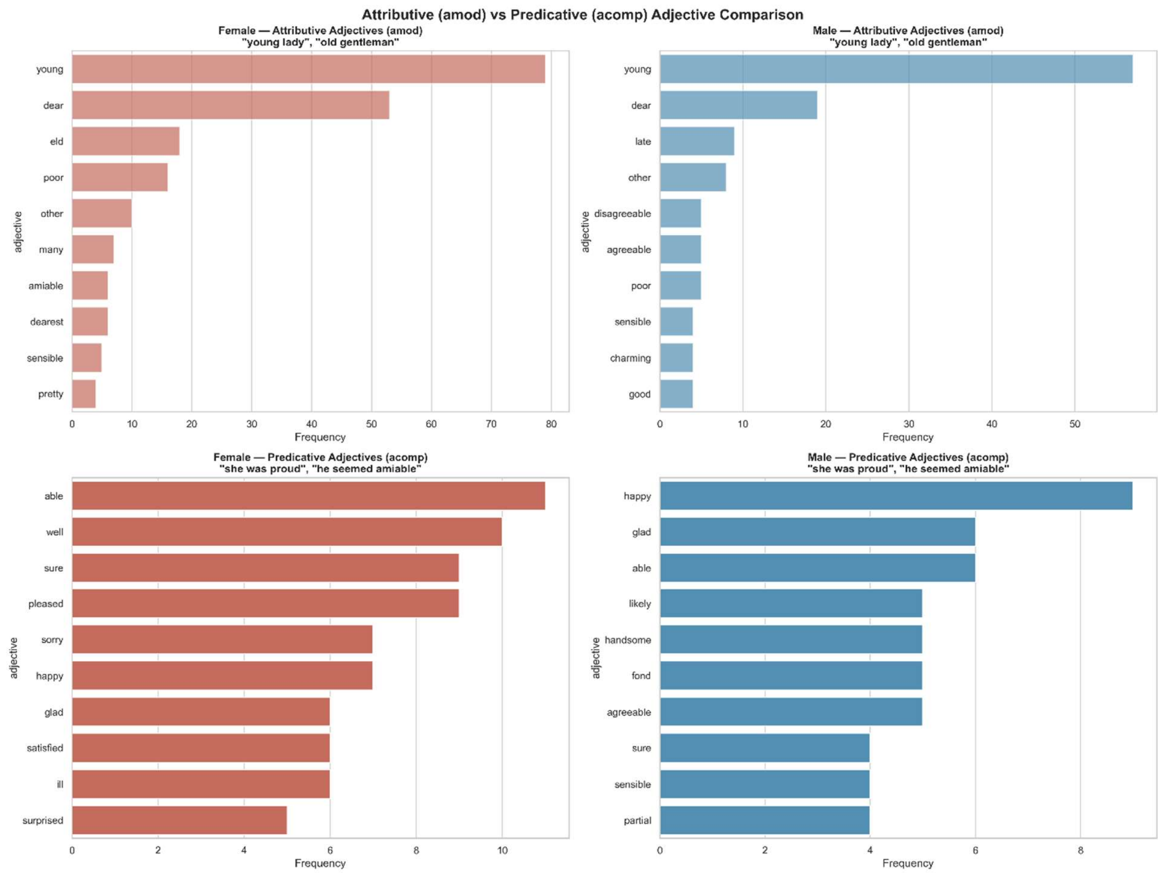

Figure 10. *Top attributive (amod) and predicative (acomp) adjectives by gender. Separating the two dependency types surfaces a richer characterisation picture than aggregate adjective counts.*

The amod/acomp separation is itself a small methodological contribution: it turns an ambiguous top-adjectives list into two interpretable columns, one for how characters are introduced and one for how they are evaluated.

5. Discussion

The findings reported in Section 4 admit a specifically linguistic interpretation. The quantitative patterns recovered from the dependency graph are, in effect, measurements of the transitivity system that Critical Discourse Analysis has long treated as a site of ideological encoding. This section reads the results through that lens. The framework applied here is van Leeuwen's (2008) social actor network — activation, passivation, relational identification — scaffolded by Fairclough's (2003) three-dimensional model of text, discursive practice, and social practice, and resting on the Hallidayan tradition (Halliday & Matthiessen, 2014) as inherited into literary stylistics by Burton (1982), Fowler (1986), and Simpson (1993).

5.1. Transitivity as an Ideological Diagnostic

The pipeline described in Section 3 can be read as a partial computational proxy for Hallidayan transitivity analysis, whether named as such or not. The agency ratio approximates one dimension of the transitivity system by tracking whether gendered referents appear in subject-like or object-like positions. It does not exhaust the full Hallidayan distinction between process types such as material, mental, and verbal processes, but it provides a tractable computational entry point into those questions. This connects the pipeline to the unit of analysis developed in critical stylistics by Burton (1982), Simpson (1993), and Fowler (1986): the distribution of participant roles, process types, and ideological positioning in fictional prose. Reading the quantitative results through CDA is therefore not an exogenous theoretical overlay; it is a recognition of what the numbers already are.

The advantage of making this recognition explicit is interpretive: it supplies a principled vocabulary — activation, passivation, relational identification, order of discourse — for what Section 4 reports in statistical form. Without it, the results risk being read either as bare numerical facts or as critical intuitions re-expressed quantitatively. With it, the patterns become evidence of the specific discursive mechanisms by which Regency-era gender ideology is textualised in the novel.

5.2. The Marriage Frame as Naturalised Discourse

The *marry*-frame result provides the clearest ideological signal in the dataset. Female referents appear as patients in 26 out of 39 cases, whereas male referents appear as patients in only 10 out of 37 cases, corresponding to patient-position rates of approximately 66.7% and 27.0%, respectively. This asymmetry is statistically robust, with Fisher's exact test rejecting independence at $p < 0.001$ and Cramér's $V = 0.397$ indicating a large effect. Interpreted through van Leeuwen's (2008) social actor framework, this pattern suggests that female referents are systematically passivated within the *marry* frame: they are grammatically rendered as the goal of an institutional process rather than its initiating actor. In this sense, the verb frame itself encodes the Regency marriage economy at the level of argument structure. The finding is not that Austen's narrator describes women as passive in some loose thematic sense; it is that the grammatical slot of the patient in the verb marry is occupied by female referents at a rate incompatible with distributional randomness at any conventional threshold.

In Fairclough's (2003) terms, this is a case of discursive naturalisation. Ideology is most durable not when it is asserted but when it is presupposed. The patterning is so structurally consistent that no character in the novel remarks on it: to be eligible to marry, from a female position, is to be eligible to be married. From the female position, the pattern is one of recurrent patient-positioning: female referents are more often grammatically cast as the object or goal of the marriage process than as its initiating agent. What the pipeline has recovered is the order of discourse (Fairclough, 2003) of the Regency marriage market, encoded at the level of the verb's argument structure.

That the asymmetry is concentrated in marry rather than distributed evenly across the lexicon is itself analytically significant. The parallel analyses of tell, admire, and persuade (Section 4.4) recover weaker or qualitatively different patterns. This uneven distribution is consistent with Baker's (2014) argument that gendered ideology is not spread smoothly through language but concentrates at institutional chokepoints — the specific lexicogrammatical sites where social power is most visibly at stake. In Pride and Prejudice that chokepoint is, unsurprisingly, the verb of matrimony.

5.3. Identity and Evaluation in the Adjective Cascade

The separation of attributive (amod) from predicative (acomp) adjectives in Section 4.5 maps cleanly onto Fairclough's (2003) distinction between two functions of discourse: the construction of identity and the enactment of relational evaluation. Attributive modifiers integrate into the noun phrase and contribute to who a social actor is constructed as, in physical and demographic terms. Predicative modifiers occur in predicate position and register how the same social actor is judged, psychologically and evaluatively, by the narrator or by other characters.

The gendered asymmetry in the distribution of these two adjective classes is therefore not a single finding but a double one: female referents receive substantial description on both the attributive and predicative axes. This is the double axis of the gendered gaze — physical construction and psychological judgement operating in parallel — and it is consistent with D'Ignazio and Klein's (2020) framing of uneven representational depth as a feature rather than a side-effect of how textual representation is organised. The narrative grants female characters a richer interior because the evaluative frame is so dense; interior depth and evaluative surveillance emerge from overlapping grammatical patterns..

5.4. The Agency Paradox as a Reflexive Finding

The most methodologically interesting result of this study is not any single statistical finding but the reflexive one: that the aggregate agency ratio, read in isolation, inverts the direction of the canonical nineteenth-century gender agency gap, while the underlying order of discourse remains firmly in the expected direction. Surface transitivity patterns can, in other words, misidentify the discursive structure they are measuring.

This has two implications for CDA practice on literary single-text corpora. First, aggregate transitivity metrics must be triangulated with point-of-view analysis; a novel with a strong focalising protagonist will produce mental-process distributions that reflect the narrator's perspectival anchoring at least as much as the surrounding social discourse. Second, frame-specific analysis is a more sensitive probe of ideological encoding than corpus-wide aggregation, precisely because it isolates the lexicogrammatical sites where the order of discourse is most densely inscribed. The agency paradox is thus not a finding that undermines Stuhler's (2024) large-corpus result but a methodological caution against reading its inverse at single-novel scale as evidence of contrary representation.

Read in this way, the three empirical findings — the marry-frame asymmetry, the attributive/predicative adjective split, and the agency paradox itself — do not merely describe Pride and Prejudice. They describe a discursive formation. The grammar of the novel reproduces the order of discourse of Regency femininity: women are activated in the domain of perception and cognition, because the narrative is anchored in Elizabeth Bennet's consciousness; and passivated in the institutional domains of marriage, social placement, and evaluation, because that is how the surrounding social practice is organised. The novel is simultaneously a site of that order of discourse and, through its protagonist's perceptual dominance, a very quiet interrogation of it.

6. Conclusion

Read in isolation, the headline agency ratio of Pride and Prejudice looks like a small victory for its female characters: women act, syntactically, more often than they are acted upon. Read against the decomposition in Sections 4.2–4.3, that victory dissolves. The higher female ratio is largely attributable to narrative focalisation: Elizabeth Bennet is the novel's perceiving and reflecting centre, and verbs of perception accumulate around her. Once attention is restricted to the marriage frame, the underlying gendered structure reappears with high statistical confidence and a large effect size. This is the study's central analytical contribution, and this study terms it the agency paradox..

Three methodological conclusions follow. First, aggregate agency ratios must be read against narrative structure at the single-text scale; they cannot be treated as standalone evidence of gendered representation when a novel has a strong focalising protagonist. Second, semantic-frame analysis of targeted verbs is a more sensitive probe than aggregate agent/patient counts, and Fisher's exact / Cramér's V is a suitable pairing for the sparse contingency tables that single-text frames produce. Third, separating attributive from predicative adjectives consistently surfaces information that aggregate adjective counts hide. Taken together, and as developed in Section 5, these observations also function as a methodological caution for Critical Discourse Analysis on literary single-text corpora: surface transitivity patterns can misidentify the underlying order of discourse unless triangulated with narrative point of view.

Three scholarly conclusions follow. The study is consistent with Underwood, Bamman, and Lee's (2018) observation that gender divisions are sharpest in the mid-19th century by showing that the underlying structure persists even when surface metrics invert. It complicates Stuhler's (2024) large-corpus agency gap by demonstrating that narrative focalisation can reverse its direction at single-novel scale without reversing its substantive content. And it aligns with Baker's (2014) case for treating collocational patterns as evidence of ideological structure, the marry frame is precisely such a pattern, and precisely such a piece of evidence.

The study's scope is deliberately narrow: one novel, one author, one period, one language. Generalisation to Austen's other novels, and to the wider 19th-century canon, is proposed as the natural next step. A further extension would be FlexiConc (Dykes et al., 2026), a Python package designed to assist concordance reading and systematic concordance analysis, which could replace the bespoke collocation component with a more reusable tool.

7. Contributions and Significance

The revised study makes three contributions. First, it applies macro-level distant-reading methodologies — developed and validated on corpora of thousands of novels — to a single canonical text, testing whether patterns identified at scale (Stuhler, 2024; Jockers & Kirilloff, 2016; Underwood et al., 2018) hold at the micro level of an individual novel. Second, it identifies an "agency paradox" in which aggregate agency ratios invert under narrative focalisation, arguing that headline agency metrics must be read against narrative structure rather than treated as standalone evidence of representation. Third, it shows that the gendered structure of Pride and Prejudice is most clearly encoded not in aggregate agent/patient counts but in specific semantic frames, most prominently the marriage frame, where the patient-position asymmetry is both statistically significant and of large effect size.

The project is grounded in feminist digital humanities (D'Ignazio & Klein, 2020), treats computational findings as signposts for close reading (Mandell, 2019) rather than endpoints, and maintains transparency about the tensions inherent in applying 21st-century NLP tools to early 19th-century prose (Risam, 2018; Rhody, 2016).

8. Tools and Resources

The project is implemented in Python using spaCy (en_core_web_trf) for dependency parsing and NLP, pandas for data manipulation, SciPy for Fisher's exact testing and effect-size computation, and matplotlib/seaborn for visualisation. The NLTK/VADER and HuggingFace Transformers dependencies from the initial proposal have been removed together with the sentiment-analysis component. The primary text is sourced from Project Gutenberg (EBook #1342). Code, outputs, and documentation are maintained in a version-controlled repository.

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